

FitBuddy

Phase 2: Requirement Analysis

Smart Fitness and Wellness Application

1. Purpose

This document identifies what FitBuddy must do and the conditions under which it must operate, based on user needs collected during ideation.

2. Data to Collect from Users

- Age, gender, height, and weight
- Fitness goal (weight loss, muscle gain, maintenance)
- Activity level (sedentary, moderate, active)
- Preferences (home/gym workouts, diet type)

3. Functional Requirements

ID	Requirement	Priority
FR1	User registration, login, and profile management	High
FR2	Create and update fitness goals	High
FR3	Generate workout recommendations from user data	High
FR4	Provide nutrition and healthy lifestyle guidance	Medium
FR5	Record daily activities (type, duration, calories)	High
FR6	Display progress reports and charts	High
FR7	BMI and calorie calculation	Medium
FR8	Reminders/notifications	Low

4. Non-Functional Requirements

Attribute	Description
Usability	Clean, easy-to-navigate interface
Performance	Pages and recommendations load within a few seconds
Security	Password hashing, input validation, protected user data

Attribute	Description
Reliability	Data stored safely; minimal downtime
Scalability	Database and backend can support growing users
Compatibility	Works on modern browsers and screen sizes

5. Software Requirements

Component	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python (Flask/Django)
Database	MySQL
Tools	VS Code, MySQL Workbench, Git, Web browser

6. Hardware Requirements

- Processor: Intel i3 or equivalent (minimum)
- RAM: 4 GB minimum (8 GB recommended)
- Storage: 500 MB free space for application and database
- Internet connection for deployment and updates

7. Constraints and Assumptions

Recommendations are general wellness guidance and not a substitute for medical advice. Users are assumed to enter accurate personal details.